

CS3334 - Data Structures

Lab 1

Outline

- Simple Exercises about Linked list and Stack
- CS Online Judge – User Guidance
- Two OJ problems

Simple Exercises

1. The stack of integers $\{41, 8\}$ undergoes the following operations, in the following order: "*pop*", "*push 2*", "*push 15*", "*pop*".

What does the stack look like after all these operations have been performed? Assume the *left-most* element is the top element of the stack.

2. Use a stack to evaluate the postfix expression $2\ 6\ 1\ -\ *\ 8\ 4\ /\ -\ 3\ *$.

What is the result?

Post your answers on Canvas/Discussions!

Exercise 1 Manipulate List

Given a singly linked list, complete a function *insert(i, d)* which inserts a new node with data *d* at position *i* of a Singly Linked List.

For example, "7 5 3 1" with *i*=1, *d*=2 will become -> "7 2 5 3 1 ")

If *i* is larger than the current list size, we do nothing.

```
void List::insert(int i, int d)
{
    ...
}
```

```
class ListNode
{
public:
    ListNode( int );
    ListNode( int, ListNode *);
    ListNode *get_Next()
    void set_Next(ListNode *);

private:
    int data;
    ListNode *next;
};
```

```
class List
{
public:
    List( String );
    List();
    void insert(int i, int d);

private:
    ListNode *first;
    string name;
}
```

Exercise 2 Manipulate List

Given a singly linked list, complete a function

reverse(i, j) which reverses elements from i-th element to j-th element(i,j inclusive).

For example, "0 1 2 3 4 5" with i=1, j=3 will become -> "0 3 2 1 4 5")

```
void List::reverse(int i, int j)
{
    ...
}
```

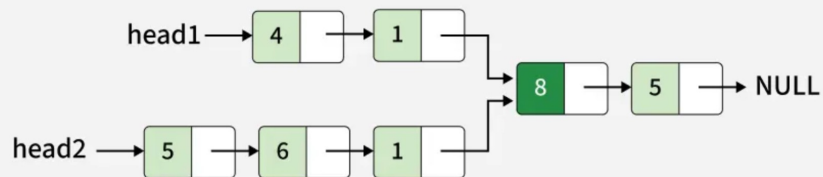
```
class ListNode
{
public:
    ListNode( int );
    ListNode( int, ListNode *);
    ListNode *get_Next();
    void set_Next(ListNode *);
private:
    int data;
    ListNode *next;
};
```

```
class List
{
public:
    List( String );
    List();
    void reverse(int i, int j);
private:
    ListNode *first;
    string name;
}
```

Exercise 3 Intersection point of two Lists

Given the head of two singly linked lists that merge at some point to form a Y-shaped structure.

The two lists may have different lengths and contain distinct nodes initially, but from the intersection point onward, they share the same sequence of nodes. Find the node at which the two linked lists first intersect.



Output is Node 8

```
Node *intersectPoint(Node
*first1, Node *first2)
{
    ...
}
```

```
class Node
{
public:
    Node( int );
    Node( int, Node *);
    Node *get_Next();
    void set_Next(Node *);
private:
    int data;
    Node *next;
};
```

```
class List
{
public:
    List( String );
    List();
    void reverse(int i, int j);
private:
    Node *first;
    string name;
}
```

Exercise 4 Stack

Given a stack, complete a function

`delete(d)` which removes all occurrences of an item d in a stack.

For example, “5 2 3 5 4” (4 is on top) with $d = 5$ will become -> “2 3 4” (4 is on top)

```
// Stack.h
#include "stdlib.h"
{
    public class Stack
    {
        public:
            Stack();
            bool IsEmpty();
            bool IsFull();
            void push(int);
            int pop();
            int top();

        private:
            ...
            // maybe array or linked based implementation

    };
}
```

```
void Stack::delete(int d)
{
    ...
}
```

Access

- <https://oj.cs.cityu.edu.hk/>
- User Account
 - *Need activation for your account. **The account activation email** has sent to your registration email*
 - *contact TA Miss Mengyang MA if you have not received email*
(provide your student id and email address)
- Please contact our TAs if you have any questions about debugging.

Problem Set

[Home](#)[Contest](#)[Record](#)[CS3391](#)[CS6382](#)[CS3334](#)[HKSC](#)

CS3334 Data Structures

(2026/27 Semester A)

Contest

12

Sample Contest

For practice only

2026-9



⌚ 20000 hour(s)

👤 0

12

CS3334-26fall-problem set

This is the problem set you need to complete.

2026-9



⌚ 8000 hour(s)

👤 2

✓ Attended

[More >](#)

Sample Contest

Sample Contest

Live... 🏆 XCPC Start at: 2026-9-12 21:15 🕒 20000 hour(s) Host: MA Mengyang 👤 0

This is a practice contest for CS3334.

It is for practice purposes only, and submission results will not affect your final grade.

WARNING:
We will check code similarity after this contest.

Problems

Status	Last Submit At	Problem
No Submissions	-	A Sums 🕒 1000ms 📄 256MiB

Step1 attend the contest

+ Attend Contest

📊 Scoreboard
🔍 Help

Status	Live...
Rule	XCPC
Problem	1
Start at	2026-9-12 21:15
End at	2028-12-24 5:15
Duration	20000 hour(s)

Step2 check problem list

☰ Problem List

🗑 End Contest Early
📊 Scoreboard
🚩 My Submissions
🔍 Help

Step3 select a problem

Example problem - Sums

Description

Find the sum of two integers.

Format

Input

Input file contains multiple test cases. Each test case consists of two integers within one line. The absolute value of each input integer is in the range $[0, 1024]$.

Output

For each test case, print the sum in separate lines.

Samples

Sample Input 1

```
0 0
1 1
```

[Copy](#) [Fill to Pretest](#)

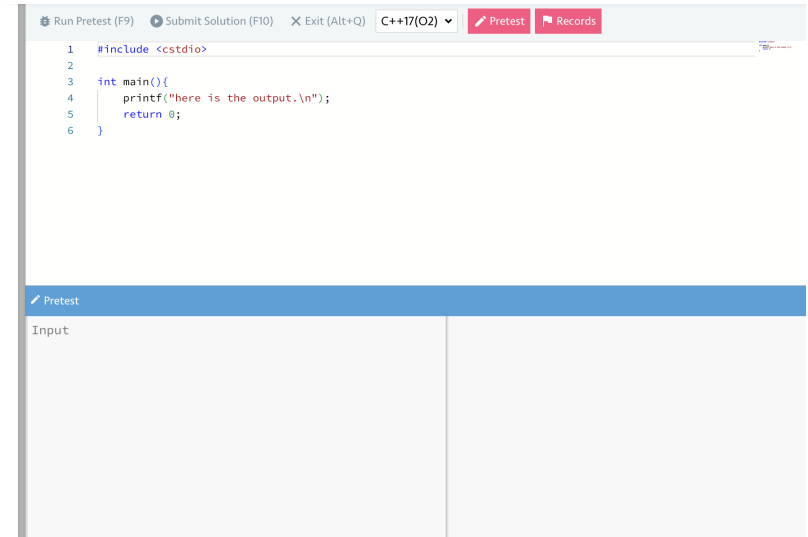
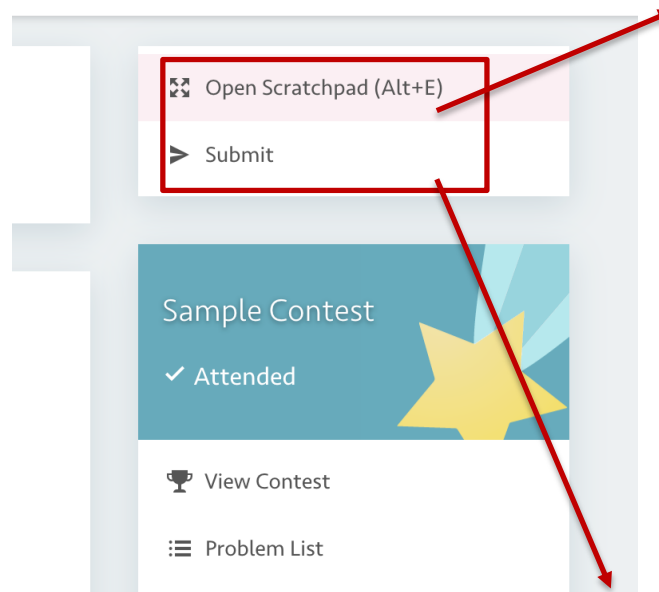
Sample Output 1

```
0
2
```

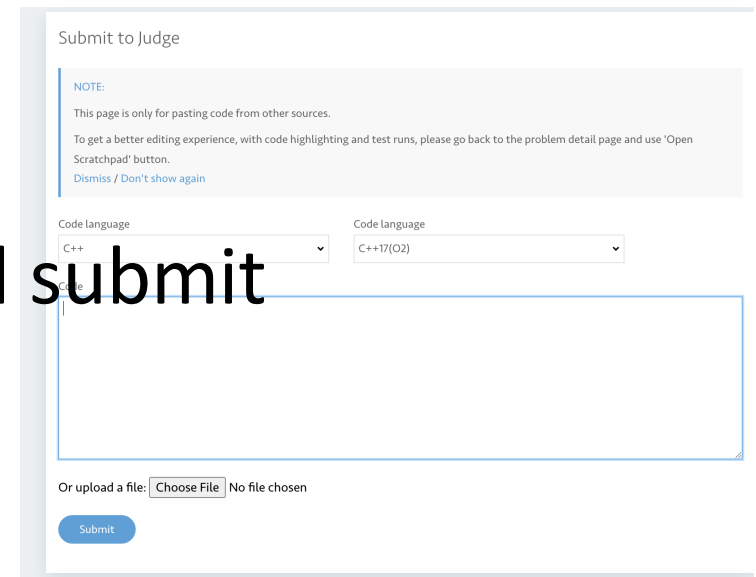
[Copy](#)

Example problem - Sums

Option 1. Code online



Option 2. Code locally and submit



Option 1. Code online

Step 1: code and test

Run Pretest (F9) Submit Solution (F10) Exit (Alt+Q) C++17(O2) Pretest Records

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int a,b;
6     while( cin >> a >> b ) {
7         cout << a+b << endl;
8     }
9     return 0;
10 }
11
```

Coding space

Pretest

0 0	Personal test case	Accepted 1.053ms 256KiB
1 1		

output

0
2

Step 2: submit

Run Pretest (F9) Submit Solution (F10) Exit (Alt+Q) C++17(O2) Pretest Records

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int a,b;
6     while( cin >> a >> b ) {
7         cout << a+b << endl;
8     }
9     return 0;
10 }
11
```

Pretest

0 0		Accepted 1.053ms 256KiB
1 1		

Records

✓ Accepted

Submission result

Option 2. Code locally and submit

Submit to Judge

NOTE:

This page is only for pasting code from other sources.

To get a better editing experience, with code highlighting and test runs, please go back to the problem detail page and use 'Open Scratchpad' button.

[Dismiss](#) / [Don't show again](#)

Code language

C++

Code language

C++17(O2)

Code

|

Step 1. Paste your code

Or upload a file: No file chosen

Step 2. submit

Submit

Verdict Information

1. Before execution:

- ❑ **Fetches**: your submission code is in the queue, to be compiled.
- ❑ **Compile Error (CE)**: The server could not **compile** your submission code. Of course, warning messages are not error messages.

mostly it is caused by [syntax errors](#):

- \$ variable not defined,
- \$ variable type not match,
- \$ need include library

Verdict Information

2. During execution:

2.1 Program finishes:

- ☐ **Accepted (AC):** OK! Your program is correct!
- ☐ **Wrong Answer (WA):** output is not correct.
 - \$input format,
 - \$special case, any [tricky](#) test case?

2.2 Program fails to finish:

- ☐ **Runtime Error (RTE):** Your program failed during the execution.
 - \$ index overflow, $a[-1]$, $a[n+1]$.
 - \$ Stack/memory overflow, endless recursions, self call
- ☐ **Time Limit Exceeded (TLE):** Your program is not finished in required time. {[time complexity](#), [infinite loop](#), [waiting for input](#),}
- ☐ **Memory Limit Exceeded (MLE):** Your program tried to use more memory than allowed.
[static memory](#) (array size) + [memory](#) for stack/queue during execution

Jumping Mario

Description

Mario is in the final castle.

He now needs to jump over few walls and then enter the Koopa's Chamber where he has to defeat the monster in order to save the princess.

For this problem, we are only concerned with the “jumping over the wall” part.

You will be given the heights of N walls from left to right. Mario is currently standing on the first wall.

He has to jump to the adjacent walls one after another until he reaches the last one. That means, he will make $(N-1)$ jumps. A high jump is one where Mario has to jump to a taller wall, and similarly, a low jump is one where Mario has to jump to a shorter wall.

Can you find out the total number of high jumps and low jumps Mario has to make?

Jumping Mario

Input

The first line of input is an integer T ($T < 30$) that indicates the number of test cases. Each case starts with an integer N ($0 < N < 50$) that determines the number of walls. The next line gives the height of the N walls from left to right. Each height is a positive integer not exceeding 10.

Output

For each case, output the case number followed by 2 integers, total high jumps and total low jumps, respectively.

Jumping Mario

Sample Input	Sample Output
3 8 1 4 2 2 3 5 3 4 1 9 5 1 2 3 4 5	Case 1: 4 2 Case 2: 0 0 Case 3: 4 0

Hints:

1, Simple enumeration with a list?

Circle Game

Description

Sheila is a naughty girl. Yesterday she came up with a boring game and forced other classmates to play with her. The rule is simple.

Some classmates will form a circle initially. Each classmate will be identified by a tag (an integer). Unluckily, you are chosen to be the judge of this game and answer Sheila's queries.

Sheila will give the following types of orders:

1. Student with tag T is added to the circle.
2. Remove a student from the circle
3. Do the query on current circle.

Please write a class to implement linked list for this problem and do not use the libraries `<list>` `<vector>` etc.

Circle Game

Input

The input consists of a single test case. The first line contains an integer N ($1 \leq N \leq 1000$) indicating the initial number of students in the circle.

Then follows N integers indicating the tags of the N students in clockwise direction. The circle begins with the first number in the sequence. Then there is an integer Q ($1 \leq Q \leq 100$) denoting the number of operations. Initially, the reference point points to the first student, then after each operation, the reference point is updated.

There will be 3 kinds of operations in the following format:

(1) 1 i tag : A new student will be inserted after the i -th student counting from the reference point in clockwise direction. It is guaranteed that i is a positive integer. After this operation, the new reference point points to the newly inserted student.

(2) 2 i : remove the student at i -th position counting from the reference point in clockwise direction. The new reference point points to the student who follows the removed student.

(3) 3 i : print the tag of the i -th student counting from the reference point in clockwise direction. The new reference point points to this student.

It may happen that $i \geq$ (total number of students in the circle), in which case you may use modulo function to make your program faster.

Circle Game

Sample Input	Sample Output
6 1 5 3 3 2 10 4 1 3 20 3 3 2 5 3 6	2 5

Hints: